Design Calculation

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1. Assumed Material Properties (PLA Plastic)

Typical PLA properties:

Property	Value
Tensile Strength	60 MPa
Yield Strength	50 MPa
Density	1.24 g/cm³
Young's Modulus	3.5 GPa

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2. Applied Load Calculation

Step 1: Assume Human Hand Force

For a small manual plastic model, assume a person applies:

$$F_{hand} = 15\ N$$

Step 2: Lever Dimensions

Assume:

- Handle length = 300 mm
- Distance from pivot to blade = 50 mm

Mechanical Advantage:

$$MA = \frac{300}{50} = 6$$

Step 3: Output Force (Applied Load)

Formula:

$$F_{output} = F_{input} \times MA$$
$$F_{output} = 15 \times 6$$

$$F_{output} = 15 \times 6 = 90\ N$$

$$F_{output} \approx 88.3\ N$$

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3. Stress Calculation

Stress formula:

$$\sigma = \frac{F}{A}$$

Assume blade support cross-sectional area:

- Width = 20 mm
- Thickness = 10 mm

So,

$$A = 20 \times 10 = 200\ mm^2$$

Now stress:

$$\sigma = \frac{88.3}{200} = 0.4415\ MPa$$

Result

Calculated stress:

$$\sigma = 0.44\ MPa$$

This is much lower than PLA yield strength (50 MPa), so the design is safe.

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4. Factor of Safety (FOS)

Formula:

$$FOS = \frac{Yield\ Strength}{Working\ Stress}$$

$$FOS = \frac{50}{0.44}$$

$$FOS = 113.6$$

Result

Factor of Safety ≈ 113

This indicates the structure is highly safe under the applied load.

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5. Lever Mechanical Advantage

Assume:

- Handle length = 300 mm
- Blade pivot distance = 50 mm

Formula:

$$MA = \frac{300}{50} = 6$$

Result

Mechanical Advantage = 6

Meaning:

- Input human force gets amplified 6 times.

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6. Strain Calculation

Formula:

$$\epsilon = \frac{\sigma}{E}$$

Where:

- $E = 3500\ MPa$

$$\epsilon = \frac{0.44}{3500} = 1.25 \times 10^{-4}$$

Result

Very small strain value indicates minimal deformation.